

Centrifugal Pump P-102 Condition Inspection Report

Quarterly Preventive & Predictive Maintenance Survey | Unit 101 (Crude Distillation) | MRPL Mangalore

ASSET TAG / ID	EQUIPMENT DESCRIPTION	INSPECTION DATE	CONDITION STATUS
P-102 (Booster)	API 610 BB2 Crude Charge Pump	July 14, 2026 (Q3)	ATTENTION REQUIRED
OPERATING SPEED / PWR	PROCESS STREAM / UNIT	LEAD INSPECTOR / ANALYST	ISO 10816-3 CLASSIFICATION
2,980 RPM 315 kW	Unit 101 Heavy Crude Bottoms	A. Verma (ISO Cat IV #2940)	Zone C (Unacceptable Long-term)

Executive Inspection Summary:

Comprehensive Q3 physical and predictive condition monitoring of booster pump **P-102** reveals an accelerated deterioration pattern. Overall vibration velocity has climbed to **3.7 mm/s RMS**, an increase of **+76.2%** relative to the January baseline (2.1 mm/s) and directly breaching the manufacturer continuous operation threshold of **3.0 mm/s** (P-102 Operating Manual §4.2). Concurrently, drive-end bearing temperature has reached **77°C (+13.2%)** with audible bearing rumble. While imminent catastrophic seizure is **NOT** indicated, immediate mechanical realignment and lubrication overhaul are mandatory to avert unplanned tripping of Unit 101 and downstream Reactor R-101.

1.0 Chronological Sensor & Telemetry Progression

Inspection Cycle	Survey Date	Vibration RMS (mm/s)	DE Temp (°C)	NDE Temp (°C)	Discharge (bar)	Condition Assessment
Baseline (Q1)	10-Jan-2026	2.10	68.0	64.5	14.1	✓ ISO Zone A (Good)
Routine (Q2)	11-Apr-2026	2.80	71.4	67.2	13.9	■ ISO Zone B (Elevated)
Comprehensive (Q3)	14-Jul-2026	3.70 (+76.2%)	77.2 (+13.2%)	74.0 (+14.7%)	13.8	✗ ISO Zone C (Exceeded)
Advisory Limit	OEM Manual §4.2	3.00 Max	75.0 Max	75.0 Max	14.0 ± 0.5	Continuous Operational Boundary

2.0 Vibration Spectrum & FFT Diagnostic Analysis

Frequency Component	Exact Peak (Hz)	Amplitude (mm/s)	Fault Signature Diagnosis	Severity
1X Running Speed	49.67 Hz	1.82 mm/s	Dynamic unbalance; slight impeller fouling from heavy bottoms.	Moderate
2X Harmonic	99.34 Hz	1.58 mm/s	Angular / Parallel Misalignment across flexible spacer coupling.	High Alert
BPFO (Bearing Outer Race)	184.20 Hz	0.45 mm/s	Incipient outer raceway micro-spalling on Drive-End bearing.	Monitor
Blade Pass Frequency (BPF)	298.00 Hz (6-blade)	0.22 mm/s	Hydraulic vane interaction; normal recirculatory pass.	Normal

3.0 Physical & Mechanical Subsystem Checklist

Inspection Subsystem	Observed Physical Condition	Engineering Assessment	Status
Baseplate & Anchor Bolts	Holding bolts inspected; torque verified at 285 Nm. No cracks in grout.	Foundation rigid; no soft-foot detected on casing pads.	PASS
Coupling & Guard	Metallic disc-pack coupling. Minor elastomeric wear dust inside guard.	Consistent with 2X harmonic misalignment signature.	ATTENTION
Mechanical Seals (Plan 53A)	Barrier fluid reservoir level steady. Gland weepage: 1 drop/min.	Within allowable operational limit (<5 drops/min). Seal face intact.	PASS
Lube Oil (ISO VG 68)	Oil level in sight glass normal; slight darkening observed.	Viscosity 64 cSt @ 40°C. Lab oil sampling recommended.	MONITOR
Suction Strainer DP	Differential pressure across strainer: 0.18 bar (Clean).	No evidence of cavitation, cavitation noise, or starvation.	PASS

4.0 Process Topology & Operational Consequence (P&ID;)

Topological Flow Path:	Crude Buffer Drum T-101 -> Booster Pump P-102 -> Control Valve V-204 -> Hydrocracker Reactor R-101
Process Hazard Analysis:	Booster Pump P-102 provides essential feed pressure to the continuous Hydrocracker reaction train (R-101). An unscheduled trip of P-102 induces immediate feed starvation, causing severe thermal runaway risk in R-101 catalyst beds, triggering emergency flare depressurization and an estimated production downtime impact of \$280,000/day.

5.0 Corrective Action Plan & Maintenance Work Orders

WO Priority	Action Description	Target Window	Responsible Craft	Expected Outcome
P1 (Urgent)	Laser Shaft Realignment: Perform hot laser alignment across pump-motor coupling; replace worn spacer discs.	Within 72 Hours	Millwright / Reliability	Eliminate 2X harmonic; reduce vibration below 2.5 mm/s.
P2 (Routine)	Lube Oil Flush & Replenishment: Drain bearing housings, flush debris, refill with synthetic Mobil SHC 626.	Within 7 Days	Lubrication Specialist	Reduce DE bearing temp from 77°C to <70°C.
P3 (Planned)	Spectral Re-Survey & Bearing Overhaul: 2-week follow-up vibration survey. If BPFO persists, replace DE bearing.	Next Plant Window	Rotating Equip Engineer	Prevent outer ring fatigue failure.

INSPECTED & REPORTED BY:	TECHNICAL REVIEW & CONCURRENCE:	KAVACHAI SOVEREIGN AI VERIFICATION:
A. Verma Lead Vibration Analyst (ISO 18436 Cat IV) Refinery Condition Monitoring Group Signature: [Verified Digital Seal]	Dr. R. K. Nair Chief Rotating Equipment Engineer Plant Reliability Division · MRPL Approval: APPROVED FOR WORK ORDER	Verification Status: VERIFIED (95%) Telemetry Hash: sha256:7f4c029... Grounded Evidence: 100% Citations Intact Air-Gapped Sovereign Enforcement